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Professional Summary

Assistant Professor and former Research Scientist and Senior Formal Verification Engineer with expertise in **Neurosymbolic AI, AI4Code, Software Engineering, and Formal Verification**. My research bridges formal methods and generative AI to enable safe, reliable, and sustainable software evolution across industrial and cyber-physical systems. I combine academic rigor (30+ peer-reviewed publications and 3 book chapters) with industrial innovation (patents and defensive publications). My goal is to advance trustworthy AI and verifiable software engineering through impactful research and collaborations with academia and industry.

Education

- 2009–2017 **Ph.D. in Computer Science and Engineering**, Indian Institute of Technology, Kharagpur, Advisors: Prof. C. Mandal, Prof. D. Sarkar
Thesis: *Path Based Equivalence Checking of Petri net Representation of Programs for Translation Validation*
- 2004–2008 **B.Tech in Computer Science and Engineering**, West Bengal University of Technology, CGPA: 8.43/10.0

Professional Experience

- Aug 2026–present **Assistant Professor**, Ashoka University, India
Courses: Operating System Lab
- Jul 2023–Jul 2026 **Scientist**, ABB Corporate Research Center, India
Validation of Generative AI-based code generation for industrial automation systems and formal verification of PLC software migration. Contributed to the ABB Co-Pilot project on AI-driven code verification and reliability assurance.
- May 2022–Jun 2023 **Senior Formal Verification Engineer**, NVIDIA, India
Developed automated formal flows for ML-based hardware accelerators; explored LLMs for design intent validation.
- Dec 2018–May 2022 **Assistant Professor**, BITS Pilani, K.K. Birla Goa Campus, India
Taught Compiler Construction, Theory of Computation, and Parallel Computing. Supervised B.Tech/M.Tech projects on formal verification.
- Aug 2017–Oct 2018 **Postdoctoral Fellow**, Hasso Plattner Institute, Germany
Worked on model-driven engineering and multi-paradigm modeling for Cyber-Physical Systems.
- Dec 2016–Jul 2017 **Assistant Professor**, BITS Pilani, K.K. Birla Goa Campus, India
Taught Compiler Construction, Theory of Computation, and Parallel Computing. Supervised B.Tech/M.Tech projects on formal verification
- Sep 2015–Dec 2016 **Visiting Faculty**, BITS Pilani, K.K. Birla Goa Campus, India
Taught Compiler Construction, Theory of Computation, and Parallel Computing. Supervised B.Tech/M.Tech projects on formal verification

Research Vision and Interests

My research aims to build trustworthy AI-driven verification frameworks integrating formal reasoning into generative engineering workflows.

Focus Areas:

- Neurosymbolic AI
- AI4Code
- Software verification
- Formal methods
- PLC verification
- Sustainable AI and hallucination detection via formal analysis

Selected Achievements

- 2024 Finalist, Global HackSummit (ABB)
- 2024 ABB Best Employee Award
- 2021 Best Paper Award, ICSOFT
- 2021 Bronze Medal, ACM SRC Grand Final (Mentor)
- 2019 Selected among Top 50 Young Researchers, Heidelberg Laureate Forum
- 2017 HPI-Humboldt Post doctoral Research Fellowship (Somad)
- 2012 TCS Innovation Labs Research Fellowship

Publications Summary

3 Journal Papers (Acta Informatica, Parallel Processing Letters)

25+ Peer-reviewed Conference Papers (ATVA, FSE, ICSE, MODELS, APSEC, etc.)

3 Book Chapters (Elsevier Press)

2 Defensive Publications

DBLP: DBLP Profile Link

Google Scholar: Citation: 234, H-Index: 9

Journal Publications

- **S Bandyopadhyay**, D. Sarkar, C. Mandal, H. Giese, "Translation Validation of Coloured Petri Net Models of Programs on Integers," *Acta Informatica*.
- **S Bandyopadhyay**, D. Sarkar, C. Mandal, "Equivalence Checking of Petri Net Models of Programs Using Static and Dynamic Cut-points," *Acta Informatica*.
- **S Bandyopadhyay**, D. Sarkar, K. Banerjee, C.A. Mandal, K. Raju, "A Path Construction Algorithm for Translation Validation Using PRES+ Models," *Parallel Processing Letters*.

Conference and Workshop Publications

- A Mukherjee, **S Bandyopadhyay**, R Halder, D Blouin, "Don't Just Translate: Verify - LLM-Guided Solidity Migration with Semantic Guarantees", *ICSOFT 2026*
- **S Bandyopadhyay**, S. Sarkar, "Antarbhukti: Verifying Correctness of PLC Software During System Evolution," *ATVA 2025*.
- **S Bandyopadhyay**, R. Jetley, "Pn4PLC: Verification of Software Upgrade for PLC Code," *FSE Companion 2025*.
- M. T. Alam, S. Goswami, K. Singh, R. Halder, A. Maiti, **S. Bandyopadhyay**, "SolGen: Secure Smart Contract Code Generation Using Large Language Models via Masked Prompting," *ISEC 2025*.
- H. Koziolk, V. Ashiwal, **S. Bandyopadhyay**, C. K. R., "Automated Control Logic Test Case Generation Using Large Language Models," *ETFA 2024*.
- R. Mittal, D. Blouin, A. Bhohe, **S. Bandyopadhyay**, "Solving the Instance Model-View Update Problem in AADL," *MODELS 2022*.
- R. Mittal, D. Blouin, **S. Bandyopadhyay**, "PNPEq: Verification of Scheduled Conditional Behavior in Embedded Software," *APSEC 2021*.

- R. Mittal, R. Banerjee, D. Blouin, **S. Bandyopadhyay**, "Translation Validation of Thread-level Parallelizing Transformations Using Colored Petri Nets," *ICSOF 2021 (Best Paper)*.
- R. Mittal, R. Banerjee, S. Sarkar, **S. Bandyopadhyay**, "Translation Validation of Loop-involving Code Optimizing Transformations Using Petri Net Models of Programs," *Petri Nets Workshop 2020*.
- Shivam, N. Goswami, V. Baths, **S. Bandyopadhyay**, "AES: Automated Evaluation Systems for Computer Programming Courses," *ICSOF 2019*.
- **S. Bandyopadhyay**, D. Sarkar, C. Mandal, "SamaTulyataOne: A Path Based Equivalence Checker," *ISEC 2019*.
- S. Sarkar, P. Kandelwal, **S. Bandyopadhyay**, H. Giese, "Analysis of GPGPU Programs for Data-race and Barrier Divergence," *ICSOF 2018*.
- **S. Bandyopadhyay**, S. Sarkar, D. Sarkar, C. Mandal, "SamaTulyata: An Efficient Path Based Equivalence Checking Tool," *ATVA 2017*.
- **S. Bandyopadhyay**, S. Sarkar, K. Banerjee, "An End-to-End Formal Verifier for Parallel Programs," *ICSOF 2017*.
- **S. Bandyopadhyay**, K. Banerjee, "PRESGen: A Fully Automatic Equivalence Checker for Validating Optimizing and Parallelizing Transformations," *HPDC Workshop 2017*.
- **S. Bandyopadhyay**, D. Sarkar, C. Mandal, "An Efficient Path-based Equivalence Checking for Petri Net Models of Programs," *ISEC 2016*.
- **S. Bandyopadhyay**, K. Banerjee, "Implementing an Efficient Path Based Equivalence Checker for Parallel Programs," *HPDC Workshop 2016*.
- K. Banerjee, **S. Bandyopadhyay**, S. Sarkar, "Data-Race Detection: The Missing Piece for an End-to-End Semantic Equivalence Checker for Parallelizing Transformations of Array-Intensive Programs," *PLDI Workshop 2016*.
- **S. Bandyopadhyay**, D. Sarkar, C. Mandal, "Validating SPARK: High-Level Synthesis Compiler," *ISVLSI 2015*.
- **S. Bandyopadhyay**, D. Sarkar, C. Mandal, "A Path-based Equivalence Checking Method for Petri Net Models of Programs," *ICSOF-EA 2015*.
- **S. Bandyopadhyay**, D. Sarkar, C.A. Mandal, "An Efficient Equivalence Checking Method for Petri Net Models of Programs," *ICSE 2015*.
- **S. Bandyopadhyay**, K. Banerjee, D. Sarkar, C.A. Mandal, "Translation Validation for PRES+ Models of Parallel Behaviours via an FSM Equivalence Checker," *VDAT 2012*.

Poster Publications

- **S. Bandyopadhyay**, "Behavioural Verification Using Petri Net Based Models of Programs," *POPL 2015 (ACM SRC)*.
- **S. Bandyopadhyay**, D. Sarkar, C.A. Mandal, "Translation Validation Using Path-Based Equivalence Checking of Petri Net Based Models of Programs," *WEPL 2015*.

Book Chapters

- B. Tekinerdogan, R. Mittal, R. Al-Ali, M. Iacono, E. Navarro, **S. Bandyopadhyay**, K. Vanherpen, A. Barisic, "A Feature-based Ontology for Cyber Physical Systems," in *Multi-Paradigm Modelling Approaches for Cyber-Physical Systems*, Elsevier, 2020. ISBN 9780128191064.
- H. Giese, D. Blouin, R. Al-Ali, H. Mkaoua, **S. Bandyopadhyay**, M. Iacono, M. Amrani, S. Klikovits, F. Erata, "An Ontology for Multi-Paradigm Modelling," in *Multi-Paradigm Modelling Approaches for Cyber-Physical Systems*, Elsevier, 2020.
- D. Blouin, R. Al-Ali, H. Giese, S. Klikovits, **S. Bandyopadhyay**, A. Barisic, F. Erata, "An Integrated Ontology for Multi-Paradigm Modelling for Cyber-Physical Systems," in *Multi-Paradigm Modelling Approaches for Cyber-Physical Systems*, Elsevier, 2020.

Intellectual Property

- **S. Bandyopadhyay**, G.K. Pandey, R. Jetley, A. Mandal, R. Singh, J. S., "Validating Migrated Code Using Equivalence Checking," *Defensive Publication*, IP.com, May 2025.

Sponsored and Consultancy Projects

- 2020–2022 “APP-based Learning for Python Programming,” Funded by 6th Sense & AGH Advisor, INR 15.8L.
- 2021–2022 “Modeling and Verification of Bio-Inspired Systems,” DST BIO-CPS, INR 20L.
- 2018–2021 “AES: Automated Evaluation System for Programming Courses,” BITS Pilani, INR 2L.

Teaching Experience

- Courses at BITS Goa Compiler Construction, Theory of Computation, Parallel Computing, Data Structures, Programming Languages, Computer Architecture, OOPS Lab.
- Industry Teaching Roadmap of Functional Verification for ML-based Hardware Accelerators; Software Engineering for AI (ABB, NVIDIA).

Professional Activities

- Reviewer CAV, DAC, EMSOFT, Acta Informatica, ACM TOSEM.
- PC Member ICSoft (2018–2023), ISEC 2019-2026, INDICON, VLSI-D, MPM4CPS, BuildSys 2025, PERR 2026, MoDeVA (2023-2025)
- Co-chair PERR Workshop, FLoC 2022.
- Membership IEEE Senior Member, ACM Member, INSTICC Member.

Key Industrial Projects

- ABB CodeGenAI:ChatGPT for PLC/DCS Control Logic and testcase Generation
- ABB FormHAL:Hallucination Detection via Formal Logic
- ABB PE4PLC: A Formal Verifier for Software Migration from Legacy PLC Code to Modern Implementation.
- NVIDIA Grace:Data-path verification for AI accelerator systems.

Technical Skills

- Programming Python, C, C++, Verilog, Shell scripting, $\LaTeX$
- GenAI Azure OpenAI, LLaMA, Claude, Gemini, RAG, LoRA
- Domain
- Formal Tools KLEE, CBMC, JasperGold, VC Formal, SAL, CPN Tools

Tool Development

- LLMA PLC Software Upgrade Using LLMs
- ABB co-pilot Automated Control Logic Test Case Generation using LLM.
- Antarbhukti Verifier for Software evolution for PLC code
- PEQ4PLC A Verification Framework for Reliable Migration of Heritage PLC Software to Modern Platforms
- SamaTulyata Formal equivalence checker for parallel programs.

References

Available on request